American Computer Science League

2022-2023 ● Contest 4: Short Problems Solutions ● Classroom Division

1. Graph Theory

The adjacency matrix raised to the 3rd power for paths of length 3 is as follows:

$$\begin{vmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{vmatrix}^3 = \begin{vmatrix} 2 & 2 & 3 & 2 \\ 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 2 & 1 \end{vmatrix}$$

Adding the entries in the right matrix gives 26 paths of length 3.

D. 26

2. Graph Theory

There are 6 cycles from A: ABA, ADA, ADBA, AEDA, AEDBA, ABCDA

E. 6

3. Digital Electronics

The Boolean expression for the digital circuit is: $\overline{A + AB + B}$

$$\begin{aligned} \overline{A + AB + B} &= \overline{A}\,\overline{AB}\,\overline{B} \\ &= \overline{A}\,(\overline{A} + \overline{B})\,\overline{B} \\ &= \overline{A}\,\overline{B} + \overline{A}\,\overline{B} \\ &= \overline{A}\,\overline{B} \end{aligned}$$

D. $\overline{A}\,\overline{B}$

4. Digital Electronics

The following truth table represents the digital circuit:

			X	Y	Z	
A	B	C	Rect(A, B, C)	AX	XC	Y+Z
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	1	0	1	1
1	0	0	0	0	0	0
1	0	1	1	1	1	1
1	1	0	1	1	0	1
1	1	1	0	0	0	0

D. 3

5. Assembly Language

It finds the first Pell number that is a perfect square after 0 and 1.
A Pell number is a(n) = 2 * a(n-1) + a(n-2).
They are 0, 1, 2, 5, 12, 29, 70, 169, 408.

D. 169

6. Graph Theory

$$\begin{vmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \end{vmatrix}^2 = \begin{vmatrix} 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 2 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{vmatrix}$$

Summing the entries in the squared matrix gives the number of paths of length 2 which is 18.

D. 18

7. Graph Theory

The cycles not containing A are: BCB, BCEDB, and CEDC

B. 3

8. Digital Electronics

The Boolean expression represented by the digital circuit is:

$$\overline{(A + (\overline{A\overline{B}} + C)) + C}$$

$$\overline{(A + (\overline{A\overline{B}} + C)) + C} = \overline{A + \overline{A} + \overline{B} + C + C}$$

$$= \overline{1 + \overline{B} + C} = \overline{1} = 0$$

A. 0

9. Digital Electronics

The Boolean expression represented by the digital circuit is:

$$\overline{\overline{A}B\,\overline{(B \oplus C)C}}$$

$$\begin{aligned}\overline{\overline{A}B\,\overline{(B \oplus C)C}} &= \overline{\overline{A}} + \overline{B} + \overline{\overline{(B \oplus C)C}} \\ &= A + \overline{B} + (B \oplus C)C \\ &= A + \overline{B} + (B\overline{C} + \overline{B}C)C \\ &= A + \overline{B} + 0 + \overline{B}C \\ &= A + \overline{B}(1 + C) \\ &= A + \overline{B}\end{aligned}$$

If $A + \overline{B} = 0$, then $A = 0$, $\overline{B} = 0$, $C = *$
Therefore there are 2 ordered tuples that make it FALSE: (0, 1, *)

B. 2

10. Assembly Language

It takes twelve times through the loop for a power of 2 to be larger than a multiple of 100. This is the Payroll Problem of $100 per day versus $1 the first day and doubled each day after that, but any power and any multiple could be used.

C. 12